MILITARY SPECIFICATION

SUPPRESSORS, VOLTAGE-SURGE; SELENIUM TYPE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers selenium voltage-surge suppressors (cells, stacks, and assemblies). The stack is the basic item of supply. More complex circuit assemblies (see 6.4.5) can be constructed using more than one stack.

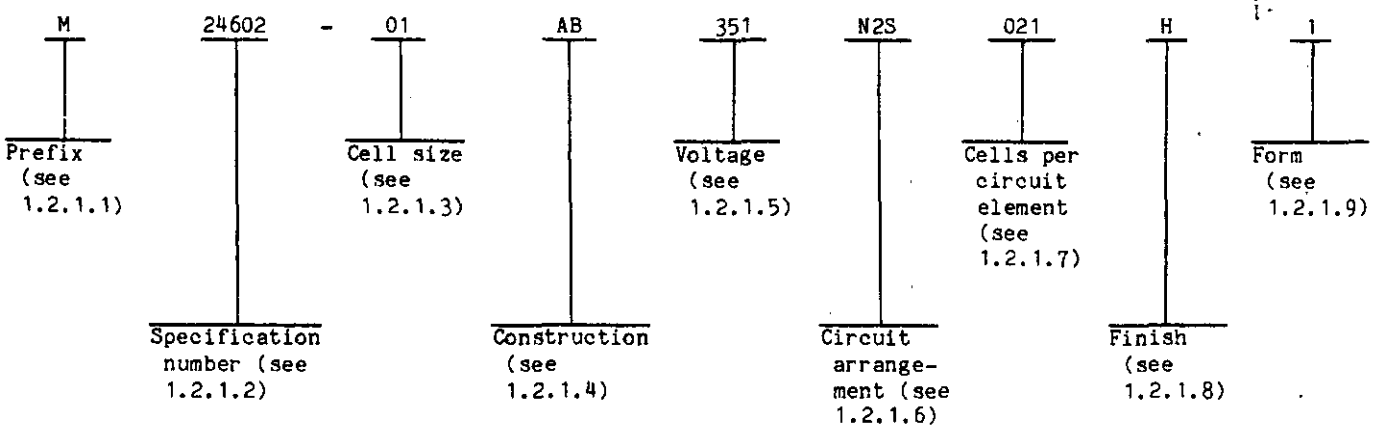
1.2 Classification. Suppressor stacks shall be of the types specified in 1.2.1 (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

1.2.1 Type designation. The type designation shall be in the following form, as specified (see 6.2.1):



The basic type designation is that portion following the dash and may be used without the prefix, specification number, and dash when available space makes necessary the use of the shorter type designation.

1.2.1.1 Prefix. The prefix M shall be used to indicate a military specification item (see 1.2.1.10).

1.2.1.2 Specification number. The basic number of this specification (24602) shall be used to designate items complying with this specification (see 1.2.1.10).

1.2.1.3 Cell size. Cell sizes shall be designated in accordance with the following coding:

Code	Cell size (inches)
01	1 x 1
02	1.25 x 1.25
03	1.6 x 1.6
04	2 x 2
05	2.2 x 2.2
06	3 x 3
07	4 x 4
08	5 x 5
09	5 x 6
10	Special (see 3.15.3)
11	5/32 diameter
12	9/32 diameter
13	15/32 diameter
14	4-3/8 diameter

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

1.2.1.4 Construction. Construction shall be designated by a two-digit code employing letters (for example AB) as shown in table I.

TABLE I. Construction designation.

Code letter	First digit, assembly type	Second digit, terminal type
A	Stud (without bracket)	Solder lugs
B	Bolt	Bolted lugs
C	Clip	Concentric leads
D	Bolt (with two brackets)	----------
E	Embedded	----------
F	Fiber case	Ferrules
H	Glass-melamine case	----------
K	Glass or ceramic case	----------
L	Stud (with one bracket)	----------
M	Metal clad	----------
R	Riveted eyelet	Radial lead
X	Special (see 3.15.3)	Special (see 3.15.3)

1.2.1.5 Voltage. The steady-state voltage rating (see 6.4.16) of the stack (direct current (d.c.) voltage for a polarized selenium surge suppressor stack (see 6.4.3) and alternating current (a.c.) root mean square (rms) voltage for a nonpolarized selenium surge suppressor stack (see 6.4.4)) corresponding to basic rating (see 6.4.25) conditions shall be designated by a three-digit number. The first two digits of this number are the first two digits of the voltage rating expressed to two significant places. The third digit is the power of 10 by which the number represented by the first two digits must be multiplied to yield the voltage rating of the stack to the nearest two significant places. For example: 200 for a polarized stack indicates 20 volts direct current (Vdc), and 352 for a nonpolarized stack indicates 3500 volts alternating current (Vac) rms.

1.2.1.6 Circuit arrangement. The circuits and connection arrangements employed in the construction of selenium surge suppressor stacks (see 6.4.2) shall be designated by a three-digit code employing letters and numerals (for example N3H) as shown in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

TABLE II. Circuit arrangement.

First digit, polarization		Second digit, number of elements	Third digit, circuits and connections	
Code	Description		Code	Description 1/
N	Nonpolarized	1	H	Half wave
P	Polarized	2	D	Doubler
		3	B	Bridge
		4	R	Ring (or delta)
		6	S	Star 2/
			X	Special 3/
C	Combination polarized and regulatory	4/		

1/ Circuit descriptions are made on the basis of their similarity to rectifier circuits. The term applies to each electrically isolated section, and there may be more than one per stack (see figure 1).

2/ The "star" circuit includes the "center-tap" and "wye" connection for 1-phase and 3-phase circuits respectively.

3/ Special circuits and connections shall be identified and documentation thereof shall be provided in the form of a specification control drawing in accordance with 3.15.3.

4/ This is a form of polarized, single element (two terminal), suppressor intended for d.c. applications which contains a specified number of cells oposing the transient flow of discharge current as from a relay coil or solenoid when it is switched off. For this circuit arrangement, only the second and third digits of this group indicate the number of cells opposing the discharge current. (The remainder of the cells (those that would perform the normal polarized suppressor function in a d.c. circuit) are indicated as specified in 1.2.1.7.)

1.2.1.7 Cells per circuit element. The number of cells in series and in parallel per circuit element shall be designated by a three-digit number. The first two digits indicate the number of cells for polarized elements or cell-pairs for nonpolarized elements in series per parallel path of each circuit element. The third digit indicates the number of parallel paths per circuit element. The product of the number represented by the first two digits and the third digit gives the total number of cells, or cell-pairs in the case of nonpolarized arrangements, per circuit element. For example: 021 for a polarized stack indicates 2 cells in series per conduction path and one conduction path per circuit element for a total of 2 cells per circuit element, and 021 for a nonpolarized stack indicates 2 cell-pairs in series per conduction path and one conduction path per circuit element for a total of 4 cells per circuit element. An exception applies for circuit arrangement coded "C" in the first digit (see 1.2.1.6) and for which the second and third digits of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

that group indicate number of cells for the regulatory function. That number must be added to the number of series cells indicated in this group and multiplied by the number of paths to obtain the total number of cells in the suppressor. For example, A03042 as these portions of the designator would indicate 3 plus 4, or 7, times 2 for 14 cells total per single element suppressor.

1.2.1.8 Finish. The finish coating of the completed stack shall be designated in accordance with the following coding:

Code	Finish coating
C	Commercial finish (see 3.13.4)
H	Military, 96-hour salt spray (see 3.13.1)
J	Potted or embedded (see 3.13.3)
M	Marine, 24-hour salt spray (see 3.13.2)
X	Special (see 3.13.5 and 3.15.3)

1.2.1.9 Form. The form, which includes features that may be peculiar to the source of manufacture as well as outline and mounting dimensions, shall be designated by the single digit number or letter assigned when qualification is granted (see 1.2.1.10).

1.2.1.10 Deviations. Primary identification of any suppressor for which deviation from full conformance to this specification has been permitted shall be by the manufacturer's FSCM code identification and applicable specification control drawing number (see 3.15.3). Secondary identification shall conform to the type designation of 1.2.1 altered as follows:

(a) The prefix (see 1.2.1.1), specification number (see 1.2.1.2), and dash shall be replaced by the word "SPECIAL".
(b) The form code (see 1.2.1.9) shall be shown as "X".

When an "X" appears in the type designator in any of the allowed positions, it provides a cue that the product must be identified by the manufacturer's FSCM code and the applicable specification control drawing number (see 3.15.3).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-79 - Plastic Rods and Tubes, Thermosetting, Laminated.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-O-16485 - Ohmmeters, Insulation Resistance-Indicating, Portable.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Suppressors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Basic requirements. Voltage surge suppressor stacks and assemblies shall conform to MIL-E-917.

3.3 Class of insulation system. Insulation system shall be class A, B, F, or H in accordance with MIL-E-917, except that for class A insulated suppressors, the electrical insulating tube over the mounting bolt or stud may be either type PBE, form Tm or type PBG, form Tr in accordance with MIL-P-79 provided this insulating tube is completely and solidly enclosed by other parts of the suppressor stack.

3.4 Insulation resistance. The insulation resistance between each electrically isolated circuit and all other circuits and ground in surge suppressor stacks and assemblies shall be not less than 100 megohms at 65°C. Conversion of values measured at other temperatures shall be made on the premise that insulation resistance doubles for each 15°C decrease in temperature.

3.5 Dielectric strength. Surge suppressor stacks shall witnstand a dielectric test of 2500 volts rms as specified in 4.5.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

3.6 Ambient temperature. The basic ratings of surge suppressor cells, stacks, and assemblies shall be based on 65°C ambient air temperature. The operating ambient temperature range is 0°C to 100°C. The storage temperature range is minus 55°C to 125°C.

3.7 Terminals (see 6.4.7) and connections.

3.7.1 Terminals for external connections. Terminals shall be provided for external connections and shall be so located as to be readily accessible.

3.7.2 Terminals with binding screws. Except as specified in 3.7.3 and 3.7.4, terminal binding screws shall be provided for all terminals for connections external to the surge suppressor stacks and assemblies. Screws shall be brass with two hexagon nuts. Screw size shall be in accordance with the following:

Suppressor cell size effective area (square inches)	Screw size
Up to 1.5	6-32
1.6 to 5.0	8-32
5.1 to 15	10-32
Over 15	1/4-28

3.7.3 Solder terminals. For surge suppressor stacks and assemblies constructed with cells or cooling fins 3 square inches or less, terminals suitable for soldering will be acceptable.

3.7.4 Wire lead terminals. For surge suppressor stacks and assemblies constructed of cells of 3/4-inch diameter or less, wire lead terminals which are coated so that they may be easily soldered with Sn 60 solder will be acceptable. The wire lead terminals shall be securely connected both electrically and mechanically to the surge suppressor and shall be at least 0.025 inch in diameter and 1.5 inches long.

3.7.5 Terminal marking. Terminals shall be marked so that the direction of surge current flow through the suppressor functioning in the suppression mode will be from positive to negative. The negative terminal shall be marked "-" and the positive terminal "+". Terminals in which the surge current may flow in either direction with the suppressor operating in its intended suppression mode shall be marked "AC". Exceptions to these basic rules for terminal marking shall be as specified in 3.7.5.1. Marking shall be by means of metal stamping, etching, or engraving and shall be located so that the marking will not be covered or obliterated by attachment of the leads. If space is not available on the terminal for marking in at least 1/16-inch block type, the marking may be placed on the body of the suppressor adjacent to the terminal. In such cases, an acceptable alternate method of terminal marking is by color coding: red for positive, black for negative, and yellow for a.c. A single method shall be employed for each product.

3.7.5.1 Exceptions to the basic rules for identification. The following exceptions shall apply to the basic rules for identification of the positive, negative, and a.c. terminals (see 6.4.8 through 6.4.10) of suppressor stacks. The nonpolarized center-tap and bridge circuits

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

(N2S, N4B, and N6B) and the polarized center-tap and star circuits (P2S and P6S) shall have their terminals marked in accordance with corresponding rectifier circuits (see 3.7.6 and figure 1). Each exception needed to best identify terminals so as to accommodate the application of suppressor stacks employing special circuits or connection arrangements shall be identified on the specification control drawing (see 3.15.3) as an exception.

3.7.6 Terminal and connection arrangements. Terminal and connection arrangement for each circuit listed on figure 1 shall be as specified therein. The arrangement for any circuit not listed on figure 1 shall be shown on the specification control drawing (see 3.15.3).

3.8 Duty. Surge suppressor stacks and assemblies shall be designed for continuous duty at their steady-state voltage rating with superimposed intermittent surge suppressor duty.

3.8.1 Storage. The surge suppressor stacks and assemblies shall be suitable for storage without any preservative treatment at temperatures from minus 20°C to 65°C, and at relative humidities throughout the range of 10 percent to 90 percent for a period of two years without deforming to the point where they will fail if inserted in a circuit without prior artificial forming (see 6.4.24). These products shall not be damaged by exposure during transit and temporary storage at temperatures throughout the range specified in 3.6.

3.9 Initial requirements.

3.9.1 Clamping voltage rating (see 6.4.17 and 6.4.18) limitation. The clamping voltage rating of a selenium surge suppressor cell (see 6.4.1), stack, or assembly shall not exceed the limit of three times its steady-state voltage (a.c. rms or d.c. average) rating.

3.9.2 Surge current and suppression voltage. The surge suppressor (cell, stack, or assembly) shall pass a nonrepetitive current of 10 amperes peak per square inch of effective cell area without its suppression voltage exceeding its clamping voltage rating and without producing "sparking" of any cell.

3.9.3 Steady-state leakage current. With rated steady-state voltage applied, the current through the surge suppressor (cell, stack, or assembly) shall not exceed 15 milliamperes per square inch of effective cell area (see 6.4.12).

3.9.4 Steady-state voltage rating limitation. The steady-state voltage (a.c. rms or d.c. average) rating of a selenium surge suppressor (cell, stack, or assembly) shall be limited to a value for which steady-state power dissipation does not exceed 500 milliwatts per square inch of effective area. The d.c. rating shall apply for sinusoidal wave shapes having a form factor (see 6.4.11) not in excess of 1.11.

3.9.5 Let-through and follow-on voltages (see 6.4.21.1 and 6.4.21.2). With any rate of rise up to 2000 volts per millisecond, and any source impedance down to 0.1 ohm per square inch per volt for the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

applied voltage surge, the suppression voltage of the suppressor (cell, stack, or assembly) shall not exceed its clamping voltage rating. For any such surge, the let-through voltage shall be no more than 10 percent greater than the maximum follow-on voltage.

3.10 Sustained operation. With a surge current (see 6.4.13) pulse of 20 microseconds (μs) duration and a 60 hertz (Hz) repetition rate which produces 0.5 ampere peak per square inch of effective area, the suppression voltage (see 6.4.21) shall not exceed the clamping voltage rating, and the steady-state leakage current shall not exceed 15 milliamperes per square inch of effective area both during and at the conclusion of 100 hours of continuous duty operation and 900 hours of intermittent operation accumulated from repetitive cycles of 5 hours on and 1 hour off (180 cycles).

3.11 Vibration. Surge suppressor stacks shall withstand the vibration test as specified in 4.5.9.

3.12 Shock. Surge suppressor stacks shall conform to grade A, class I requirements of MIL-S-901 and shall be tested in accordance with 4.5.10.

3.13 Coatings. Surge suppressor stacks shall have one of the finish coatings specified in 3.13.1 through 3.13.5, as specified (see 6.2.1).

3.13.1 Military, 96-hour salt spray. The military finish coating shall render the stack resistant to the detrimental effects of salt spray when tested as specified in 4.5.11.1 for an exposure period of 96 hours.

3.13.2 Marine, 24-hour salt spray. The marine finish coating shall render the stack resistant to the detrimental effects of salt spray when tested as specified in 4.5.11.2 for an exposure period of 24 hours.

3.13.3 Potted or embedded. The stack may be potted or embedded only if all of the following are applicable:

(a) The effective cell area is less than 0.5 square inch.
(b) The circuit and cell arrangement are such that surface-to-surface contact interfaces are not depended upon for electrical or thermal contact; or, if employed, any such contact interface is effectively sealed against the penetration of the potting or embedding compound into the interface area.
(c) Spring pressure is not relied upon for establishing or maintaining electrical or thermal contacts where the spring element can be contacted by the potting or embedding compound.

Surge suppressor stacks and assemblies which are potted or embedded shall be resistant to the detrimental effects of temperature cycling and salt spray when tested as specified in 4.5.11.3 for an exposure period of 48 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

3.13.4 Commercial finish. The commercial finish shall be that which the manufacturer employs for stacks intended for general industrial use, and the manufacturer shall identify the finish both by his catalog designation and his applicable process control specification number and date. The finish coating shall render the stack resistant to the detrimental effects of temperature-humidity cycling when tested for moisture resistance as specified in 4.5.11.4 for 10 complete cycles.

3.13.5 Special finishes. A special finish shall be required and provided only when the peculiar nature of the application environment makes one of the above-indicated finishes unsuitable. The environment, length of service, and method of test to be used to determine suitability and to assure quality shall be specified in detail. The finish coating shall be identified by its name designation and the applicable process control specification number and date. These, together with a description of the environment and the service for which the finish is designed, shall be furnished by the suppressor manufacturer on a specification control drawing (see 3.15.3) along with the test method and definition of failure to conform. The finish coating shall render the stack resistant to the detrimental effects of the specified environment when tested as specified in 4.5.11.5.

3.14 Identification plates. Identification plates shall be type A, B, or C in accordance with MIL-P-15024 and MIL-P-15024/5, except that plastic material shall not be used, and the standard dimensions do not apply. The color style selected by the contractor shall be III or IV whichever is least costly to the Government for the particular application. Identification plates shall contain the following:

(a) Manufacturer's name or assigned code symbols (see 6.5).
(b) Surge suppressor stack identification number.
(c) Tabulated detail drawing (see 3.15.2) number (or the specification control drawing number, if any of the classification categories are "special") and piece number.
(d) National Stock Number.
(e) Space for the Government inspector's stamp.

No identification plates are required for surge suppressor stacks constructed of cells 2 square inches and smaller. However, the manufacturer's name or assigned code symbols and the surge suppressor identification number (or the specification control drawing and piece number, if any of the classification categories are "special") shall be added in such a manner as will be readable for the expected life of the surge suppressor stack.

3.15 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.15.1 through 3.15.3.

3.15.1 Master drawing and supplements. Each product line that is intended for qualification under this specification shall be covered by an engineering drawing prepared in accordance with the data ordering documents (see 6.2.2). This drawing shall be identified by appropriate entry in the title block and contents placed thereon as the master drawing for the particular product line as defined by the manufacturer and produced

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

in conformance with this specification. Contents of the master drawing shall include that specified in 3.15.1.2 and such other information and data needed to support the use indicated in 3.15.1.1. Information and data shall be arranged and presented in format and detail to facilitate the intended functions and uses. The master drawing shall be supplemented by other engineering drawings or company standards as determined by the contractor to most effectively and economically comply with the requirements of the data ordering document (see 6.2.2). A data list which identifies all drawings and documents which collectively define the product line(s) for qualification purposes shall be prepared. The acronym "QPL" shall be employed in lieu of a contract number. Upon notification of qualification, the contractor shall provide a microfilm of the data list and all documents listed thereon, except military specifications, standards, and publications. Parts and materials lists, where applicable, shall be integral with the engineering drawings to which they apply.

3.15.1.1 Use. The master drawing is intended to serve the following functions and uses:

(a) Be the vehicle through which each product line, with its capabilities and limitations, is identified for qualification purposes.
(b) Provide basic design and construction detail for design evaluation and product line identification.
(c) Provide engineering data and application guidance to help assure proper selection and application by the equipment designer.
(d) Provide reference to tabulated detail drawings which depict the preferred items of supply, provide outline and mounting dimensions, and present application data applicable to individual items of supply as required by 3.15.2.

3.15.1.2 Contents. In addition to the information required by the data ordering document (see 6.2.2), master drawings shall contain the following:

(a) Manufacturer's identification of the type and process (including forming) for the surge suppressors covered by the drawing.
(b) List of the various sizes of cells and their effective cell area.
(c) Rating and characteristics of the cells including the steady-state voltage (a.c. rms and d.c. average) ratings, the sparking voltage rating (see 6.4.19 and 6.4.20), the clamping voltage rating, the discharge current rating, and the steady-state leakage current and power loss characteristics (maximum values) in a 65°C ambient temperature.
(d) Maximum and minimum operating temperature.
(e) Maximum number of cells per stack both for polarized and nonpolarized construction for each cell size for which qualification is desired.
(f) Length between mounting brackets of stacks with the maximum number of cells per stack.
(g) Curves of the rerating necessary when operated in an ambient temperature of other than 65°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

(h) Curves for each cell size of the maximum nonrepetitive surge current for surge durations up to and including 1 millisecond.

(i) Curves for each cell size of maximum power loss for steady-state operation with a maximum cell operating temperature of 100°C at steady-state voltage values over the range of 20 to 120 percent of rated value.

(j) Curves or tables, or both, sufficient to give rerating factors necessary to compensate for all selectively variable factors such as type of construction, cell arrangement (e.g., polarized or nonpolarized), cell spacing, number of cells per stack, finish coating, and cooling air velocity with all necessary instructions for properly applying these rerating factors.

(k) Curves or tables giving application factors necessary to compensate for, or provide for, aging effects and instructions for their use.

(l) Preferred and minimum acceptable reforming procedures together with criteria for determining when reforming is needed and also when satisfactory reforming has been accomplished.

(m) Sections of the typical surge suppressor stacks showing the details of each different type of construction. Parts for each type of construction shall be identified and information on each size and type to be employed throughout the range of cell and stack sizes shall be included. Where more than one design of a part may be employed, the information should be given for each design.

(n) The peak voltage per cell versus the peak surge current (see 6.4.14) through the cell for each cell size (maximum for stabilized cells throughout the operating temperature range).

(o) Curves for each cell size of the off time required between surges versus the energy content per surge in a 65°C ambient temperature (when applied at a.c. rms and d.c. average rating).

3.15.2 Tabulated detail drawing. A tabulated detail drawing shall be prepared in accordance with the data ordering document (see 6.2.2), and shall be provided to supplement the master drawing (see 3.15.1) and to identify each item of supply for which qualification is sought. Each family group of products having the same construction and processing control shall be covered by a single tabulated detail drawing. A separate tabulated detail drawing shall be provided for each different family group. Each tabulated detail drawing shall include the following:

(a) Manufacturer's name and plant address.
(b) Applicable military specification, including issue and date.
(c) Military type designations, in accordance with the applicable specification, for each of the preferred items of supply (each product for which qualification is sought).
(d) Manufacturer's type and catalog number corresponding to each product listed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

(e) Basic rating, outline, and mounting dimensions for each product listed, with dimensions keyed to an outline sketch which includes mounting detail and terminal locations.
(f) Application or service factors, and instructions for the use thereof, which relate the rating and application data of the master drawing (see 3.15.1) to each product listed.
(g) Reference to the suppressor master drawing covering the product line to which the family group belongs.
(h) Weight of each item of supply.
(i) Identification plate information.
(j) Terminal arrangement, marking, and interconnections.
(k) Identification of mounting and connection hardware provided.
(l) Spacing between cells.

3.15.3 Specification control drawings. Information and data necessary to identify an item of supply, its performance characteristics, or its application limitations that are different from or not contained on the master drawing (see 3.15.1) or tabulated detail drawing (see 3.15.2) shall be provided on a specification control drawing for that product (see 6.2.2). Each classification category designated as "special" (see 1.2.1.3, 1.2.1.4, 1.2.1.6, 1.2.1.8, and 1.2.1.10) shall be identified with applicable specifics on the specification control drawing. Likewise, all examination and test requirements and acceptance criteria which supplement that specified in this specification shall be detailed on the specification control drawing for the applicable product. In addition, the specification control drawing shall comply with the requirements of 3.15.2, except that the scope of coverage is that of a single item of supply, and qualification status will be determined by that applicable to the basic product line from which the item comes. Each applicable specification control drawing shall be furnished with the equipment drawing on which it is referenced to provide product identification whenever drawings are required for equipment using suppressors in accordance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Periodic tests (see 4.4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Qualification inspection shall consist of the examination and tests specified in tables III, IV, and V, as applicable. For purposes of qualifying a series of products of a particular family or product line, sampling shall be as specified in 4.3.1 with qualification inspection in accordance with 4.3.2 and 4.3.3. For purposes of qualifying individual suppressors independently of a series or product line, sampling and inspection shall be as specified in 4.3.4.

4.3.1 Sampling for qualification inspection (for series or family line). For each cell size to be tested, a sample of stacks composed as follows shall be subjected to qualification inspection in the order and manner described herein. Consider the assembly types (see 1.2.1.4) grouped and listed in the following order:

(a) A, B, R
(b) C
(c) F, H, K, M
(d) E
(e) X

For the first listed assembly type of those that are to be qualified (in each group) for the particular cell size, provide ten specimens of stacks with six cells per stack in the P6S (see 1.2.1.6) circuit arrangement. In addition, for each assembly type of those that are to be qualified for the particular cell size, provide two specimens of stacks with the maximum number of cells that are to be used in that assembly type. The NXH (see 1.2.1.6) circuit arrangement, where X is the maximum number of cells to be used per stack divided by two, shall be used for these stacks. Stud and bolt assembly types (see 1.2.1.4), when applicable, shall be furnished with mounting bracket(s) assembled in the six-cell stacks and without mounting brackets for the stacks with the maximum cells per stack.

4.3.2 Specimens with six cells per stack. The specimens with six cells per stack shall be subjected to the examination and tests specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

TABLE III. Qualification inspection on specimens with six cells per stack.

Number of specimens	Inspection	Inspection paragraph
10	Visual and mechanical examination Steady-state leakage current Surge current Insulation resistance Dielectric	4.5.2 4.5.4 4.5.5 4.5.6 4.5.7
1	Creepage and clearance distances	4.5.8
2	Vibration Shock Protective finish	4.5.9 4.5.10 4.5.11
3	Sustained operation	4.5.12
4	Rate of voltage rise Thermal stability	4.5.13 4.5.3

4.3.3 Specimens with the maximum number of cells per stack. The specimens with the maximum number of cells per stack shall be subjected to the examination and tests specified in table IV.

TABLE IV. Qualification inspection on stacks with maximum number of cells per stack.

Inspection	Inspection paragraph
Visual and mechanical examination Steady-state current Surge current Insulation resistance Dielectric Vibration Shock Protective finish	4.5.2 4.5.4 4.5.5 4.5.6 4.5.7 4.5.9 4,5.10 4.5.11

4.3.4 Sampling and qualification inspection (for individual suppressors). Offerers may qualify individual suppressors in lieu of a series or family line. In such cases, a sample of each different suppressor shall be subjected to the qualification inspection in accordance with table V. A sample shall consist of two specimens, or, at the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

offerer's option, one specimen may be subjected to all of the tests of table V. Examination and test shall be conducted in the order listed in table V. Specimens subjected to qualification inspection are not acceptable for delivery as part of a production lot.

TABLE V. Qualification inspection on specimens representing individual suppressors.

Specimen		Inspection	Inspection paragraph
Number 1	Number 2		
X	X	Visual and mechanical examination	4.5.2
X	X	Steady-state leakage current	4.5.4
X	X	Surge current	4.5.5
X	X	Insulation resistance	4.5.6
X	X	Dielectric	4.5.7
X	---	Rate of voltage rise	4.5.13
X	---	Thermal stability	4.5.3
X	---	Sustained operation	4.5.12
---	X	Vibration	4.5.9
---	X	Shock	4.5.10
---	X	Protective finish	4.5.11
---	X	Creepage and clearance distance	4.5.8

4.3.5 Data required with request for authorization of qualification tests. In submitting requests for authorization of qualification tests, the contractor shall include the following information for use by NAVSEA in selecting the sizes of cells to be tested and in establishing qualification:

(a) Three prints of a master drawing as specified in 3.15.1 for each type and process (including forming) for which qualification is desired and three prints of each tabulated detail drawing as specified in 3.15.2 for each family group of products for which qualification is desired.

(b) Description of the methods to be employed in conducting the inspection tests in accordance with the option of 4.5.1.1, evidence of correlation of the acceptance criteria under the test conditions with the specified requirements, and certification of compliance with the requirements of 4.5.1.1 together with substantiating evidence.

4.4 Quality conformance inspection.

4.4.1 Lot. Identical surge suppressors offered for delivery at one time shall be considered a lot.

4.4.2 Sampling for quality conformance inspection. Statistical sampling and inspection shall be in accordance with MIL-STD-105. The acceptable quality level (AQL) shall be 1.0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.4.3 Inspection. Each specimen of each sample selected in accordance with 4.4.2 shall be subjected to the examination and tests specified in table VI. Each failure to comply with an applicable requirement, regardless of whether said requirement is contained in the contract or order or in this specification, either directly stated or invoked through references, shall be classified as a major defect. For the purposes of applying MIL-STD-105, all forms of defects are classified as major defects.

TABLE VI. Quality conformance inspection and periodic tests.

Examination and tests	Applicable test paragraph	
	Quality conformance inspection	Periodic tests
Visual and mechanical examination	4.5.2	4.5.2
Thermal stability	-----	4.5.3
Steady-state leakage current	4.5.4	4.5.4
Surge current	4.5.5	4.5.5
Insulation resistance	4.5.6	4.5.6
Dielectric	4.5.7	4.5.7
Protective finish	-----	4.5.11
Creepage and clearance distances	-----	4.5.8
Special	When required (see 6.2.1)	------

4.4.4 Periodic tests. Two surge suppressor stacks from the first production run of each cell size, process, and assembly type of surge suppressors offered for delivery shall be subjected to the periodic tests specified in table VI. Thereafter, two surge suppressor stacks from the first production run of each cell size and process offered for delivery in the calendar year shall be subjected to these periodic tests. The specimens to be used for periodic tests shall be constructed as 2-cell stacks in the P2S (see 1.2.1.6) circuit arrangement. The assembly type for the annual test specimens shall be the same as the first production run of stacks of the particular cell size and process offered for delivery in the calendar year. In case there is more than one assembly type for a given cell size and process being offered at the same time, the first listed applicable one in 4.3.1 shall be chose for the periodic test specimens. However, on each fifth year after the year of first qualification, the periodic tests shall be conducted on test specimens representing each cell size, process, and assembly type that the contractor desires to be retained on the qualified products list. Failure to submit, by the end of each fifth year, test report(s) (see 6.2.2) showing compliance of the test specimens with the requirements of this specification shall be sufficient reason for removing the products so affected from the qualified products list. Suppressor cells and assembly parts for the construction of test specimens for all periodic tests shall be randomly chosen from production line stock by an authorized Government representative and shall be assembled in normal manner on the production line under the Government representative's scrutiny. If either of the two periodic

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

test specimens should fail any of the periodic tests, further lots containing products represented by these test specimens shall not be accepted for tests until the contractor has taken corrective measures both for future production and on all unshipped suppressor stacks, and has demonstrated to the Government representative that these measures will enable the suppressor stacks to conform to this specification.

4.5 Tests. The contractor shall demonstrate to the satisfaction of the Government representative that the test methods employed in the tests give the correct steady-state and dynamic characteristics of the surge suppressor, and that the probable error in measured characteristics does not exceed 5 percent. Conditioning and monitoring of electrical performance, when required, during mechanical and environmental tests shall be imposed simultaneously on all circuit elements of the test specimen. Performance tests and measurements, which by their nature are not required to be made on all circuit elements simultaneously, may be applied successively to the individual circuit elements of each test specimen. In every case, all circuit elements of each specimen shall be tested, and resulting test data shall be identified to the applicable circuit element and specimen.

4.5.1 Test temperature. For qualification and periodic tests, 4.5.3, 4.5.5 and 4.5.12 shall be conducted in an ambient temperature of 65°C ± 2°C. For other testing, 4.5.1.1 shall apply. Ambient temperature measurement shall be in accordance with 4.5.1.2.

4.5.1.1 Contractor's option. For those examinations and tests for which ambient temperature is not specified, the contractor or his agent performing the examination or test may employ, at the suppressor contractor's option, either room ambient temperature or 65°C ± 2°C. If the former is chosen, except for visual and mechanical examination, it shall be the responsibility of the contractor to first submit evidence satisfactory to the Government representative of correlation of acceptance criteria to be used with that specified and applicable at an ambient temperature of 65°C ± 2°C. The contractor shall certify that the acceptance criteria which applies at the test temperature is no less severe than that specified for 65°C ± 2°C and is at least as effective in determining quality of the product.

4.5.1.2 Measurement of ambient temperature. In order to avoid errors due to such factors as time lag, air stratification, and drafts, the ambient temperature shall be measured at different points around and on a level with the suppressor stack. The thermometers shall be inserted in oil-filled cups not less than 1-inch in external diameter and 2-inches high and shall be protected from drafts and from abnormal heat radiation. The value to be used for ambient temperature shall be the mean of the readings of the several thermometers placed as stated. The conditions in the testing enclosure shall be such that the ambient temperature will not vary more than 2°C during the tests.

4.5.2 Visual and mechanical examination. Each specimen of the sample of surge suppressor stacks shall be subjected to a thorough examination to ascertain that the material, workmanship, and design are in conformance with this specification, the applicable item drawing, and the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.5.3 Thermal stability. With a surge voltage of 1 millisecond duration and a 60 Hz repetition rate which produces 1 ampere peak per square inch of effective cell area, the suppression voltage on the suppressor shall not exceed its clamping voltage rating during and after 1 hour of operation. Temperatures shall be measured in accordance with method 1 of MIL-E-917. Temperature rise of hottest spot above ambient temperature shall not exceed 15°C.

4.5.4 Steady-state leakage current (see 6.4.15). The steady-state leakage current shall be measured 1 minute after application of rated steady-state voltage, and repeated at 5 minute intervals until thermal stability is achieved, except that for quality conformance inspection, readings at 1 minute shall suffice. Polarized units are to be tested with a d.c. voltage having a form factor not exceeding 1.11. The test specimen shall be in thermal equilibrium at the applicable ambient temperature immediately prior to application of the test voltage. The applicable ambient temperatures for this test are 0°C ± 2°C, 30°C ± 2°C, 65°C ± 2°C, and 100°C ± 2°C, except for quality conformance inspection and when steady-state leakage current is measured as initial and end-point measurements associated with another test (see 4.5.1.1). The steady-state leakage current shall conform to 3.9.3. Temperature of the ambient air and suppressor hottest spot shall be measured at the same times as the steady-state leakage current. Temperature rise of the hottest spot above ambient air shall not exceed 15°C. Measurement of suppressor temperature is not required for quality conformance inspection.

4.5.5 Surge current. A surge voltage pulse of 300 μs duration which produces 10 amperes peak per square inch of the effective cell area shall be impressed upon the suppressor under test. Performance shall be monitored and documented by recording the instantaneous voltage-current characteristic of the suppressor displayed on an oscilloscope. The test shall consist of a chain of 25 surge voltage pulses sufficiently spaced to avoid build-up of cell operating temperature in the suppressor. Both the suppressor and the recordings of instantaneous voltage-current characteristic shall be examined for any evidence of sparking or other malfunctioning. Any such evidence of failure to comply in any respect with 3.9.2 shall constitute failure of the specimen under the surge current test.

4.5.6 Insulation resistance. Insulation resistance shall be measured with an insulation-resistance-indicating meter conforming to type GM of MIL-O-16485, or with one of equivalent voltage and accuracy at the specified insulation resistance. The insulation resistance shall be measured from the terminals of each electrically isolated circuit to ground and all other circuits connected together. The insulation resistance shall be in accordance with 3.4.

4.5.7 Dielectric tests.

4.5.7.1 General. The dielectric test shall be made from the terminals of each electrically isolated circuit to ground and all other circuits connected together. If the insulation resistance is known to be lower than that specified due to dirt, moisture, or damage, this shall be remedied before application of the dielectric test voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.5.7.2 Test voltage. The frequency of the testing voltage shall be not less than 60 Hz and shall approximate a true sine wave. The rms value of the test voltage shall be as specified in 4.5.7.2.1 and 4.5.7.2.2.

4.5.7.2.1 For qualification and periodic tests, the designated specimen from the sample of suppressor stacks shall withstand for a period of 1 minute a dielectric test voltage of 2,500 volts rms.

4.5.7.2.2 For quality conformance inspection each specimen of the sample of suppressor stacks selected for test shall withstand for a period of 1 minute a dielectric test voltage as follows:

Voltage rating of stack (a.c. rms or d.c. average)	Test voltage rms
50 and under	500
51 to 90	800
Over 90	1000 plus twice rated

If desired by the contractor, the dielectric test may be made at a voltage of 120 percent of the specified voltage for a period of 1 second in lieu of 1 minute.

4.5.8 Creepage and clearance distances. Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with MIL-E-917.

4.5.9 Vibration. Suppressors shall be tested in accordance with type I of MIL-STD-167-1. The frequency range of the test shall be 4 to 50 Hz.

4.5.9.1 Failure on vibration. The suppressor stacks shall be considered to have failed the vibration test if there is a loosening of any part, cracking of the protective coating, the suppression voltage does not conform to 3.9.2, the steady-state leakage current does not conform to 3.9.3, the insulation resistance does not conform to 3.4, or the dielectric test at 1650 volts rms produces a breakdown of the insulation.

4.5.10 High-impact shock. The tests shall be as prescribed for type C of MIL-S-901. The shock test shall be conducted after completion of the vibration test.

4.5.10.1 Method of mounting the suppressor on the shock-testing machine. The mounting adaptor identified as fixture 6E of MIL-S-901 shall be used. The suppressor stacks shall be mounted on the applicable mounting bracket shown on figure 2 with the mounting stud parallel to the immediate supporting channels of the mounting adaptor except for suppressors with wire lead terminals. Suppressor stacks with wire lead terminal shall be mounted by soldering the wire lead terminals to stud terminals on the mounting adaptor. The stud terminals shall be of sufficient size so that there is no permanent deformation under shock. When mounted, there shall be 0.5 inch (approximately) of lead length between the suppressor case and the stud terminals, and the clearance between the suppressor case and the mounting adaptor shall be not less than 0.5 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.5.10.2 Failure on shock test. The suppressor stacks shall be considered to have failed the shock test if there is loosening of any part, cracking of the protective coating, the suppression voltage does not conform to 3.9.2, the steady-state leakage current does not conform to 3.9.3, insulation resistance does not conform to 3.4, or the dielectric test at 1650 volts rms produces a breakdown in the insulation.

4.5.11 Protective finish. The applicable protective finish test shall be selected in accordance with the designated finish coating (see 1.2.1.8) and as specified in table VII, and shall be conducted in accordance with the indicated test.

TABLE VII. Protective finish tests.

Finish coating		Requirement paragraph	Test paragraph
Code	Description		
C	Commercial	3.13.4	4.5.11.4
H	Military, 96-hour salt spray	3.13.1	4.5.11.1
J	Potted or embedded	3.13.3	4.5.11.3
M	Marine, 24-hour salt spray	3.13.2	4.5.11.2
X	Special	3.13.5	4.5.11.5

4.5.11.1 Salt spray test, 96-hour. Military, 96-hour salt spray finish coating shall be tested in accordance with method 101 of MIL-STD-202. The salt-solution concentration shall be 5 percent, and the length of the test shall be 96 hours. After exposure, each test specimen shall be cleaned of salt residue, and examined and tested as specified in 4.5.11.7. Failure is defined in 4.5.11.8.

4.5.11.2 Salt spray test, 24-hour. Marine, 24-hour salt spray finish coating shall be tested in accordance with method 101 of MIL-STD-202, except that the length of the test shall be 24 hours. The salt-solution concentration shall be 5 percent. After exposure, each test specimen shall be cleaned of salt residue, and examined and tested as specified in 4.5.11.7. Failure is defined in 4.5.11.8.

4.5.11.3 Potting or embedment integrity test. Potted or embedded finish coating shall be tested as specified in 4.5.11.3.1 and 4.5.11.3.2, with the thermal shock test performed first, followed by the salt spray test.

4.5.11.3.1 Thermal shock. The thermal shock test shall be as specified in method 107, test condition B of MIL-STD-202, except that measurements shall be made in accordance with 4.5.11.6 prior to the first cycle and in accordance with 4.5.11.7 upon completion of the final cycle. Failure is defined in 4.5.11.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.5.11.3.2 Salt spray test, 48-hour. The salt spray test shall be as specified in method 101 of MIL-STD-202. The salt-solution concentration shall be 5 percent, and the length of the test shall be 48 hours. After exposure, each test specimen shall be cleaned of salt residue, and examined and tested as specified in 4.5.11.7. Failure is defined in 4.5.11.8.

4.5.11.4 Moisture resistance. Commercial finish coating shall be tested in accordance with method 106 of MIL-STD-202. Initial measurements shall be made in accordance with 4.5.11.6. Final measurements at high humidity shall be made in accordance with 4.5.11.4.1 and after the drying period in accordance with 4.5.11.7. Failure is defined in 4.5.11.8.

4.5.11.4.1 Final measurements at high humidity. Final measurements on each test specimen at high humidity shall include insulation resistance in accordance with 4.5.6, steady-state leakage current in accordance with 4.5.4, and insulation resistance in accordance with 4.5.6 immediately after deenergization.

4.5.11.5 Special tests. When a special finish coating is required (see 1.2.1.8 and 6.2.1) or provided, the test method of the applicable specification control drawing (see 3.13.5) shall be applied to determine the effectiveness of the special finish coating. Except as otherwise specified by the applicable specification control drawing, the initial measurements shall be made in accordance with 4.5.11.6, and the final measurements in accordance with 4.5.11.7. Failure shall be determined as specified in the specification control drawing and as specified in 4.5.11.8.

4.5.11.6 Initial measurements. Initial measurements shall be made in an ambient temperature within the range of 20°C to 35°C and shall include ambient temperature in accordance with 4.5.1.2, steady-state leakage current in accordance with 4.5.4, the instantaneous voltage-current characteristic dynamically obtained under the surge current test in accordance with 4.5.5, and insulation resistance in accordance with 4.5.6.

4.5.11.7 Final measurements. Final measurements shall be made with an effective ambient temperature which is within plus or minus 5°C of the effective ambient temperature at which the initial measurements (see 4.5.11.6) were made. Final examination and measurements shall include visual and mechanical examination in accordance with 4.5.2, ambient temperature in accordance with 4.5.1.2, steady-state leakage current in accordance with 4.5.4, the instantaneous voltage-current characteristic dynamically obtained under the surge test in accordance with 4.5.5, insulation resistance in accordance with 4.5.6, and dielectric test in accordance with 4.5.7, except that the test voltage shall be 1650 volts rms.

4.5.11.8 Failure under the protective finish test. Any test specimen shall be considered to have failed the protective finish test if there are visible signs of deterioration or corrosion, the suppression voltage does not conform to 3.9.2, the steady-state leakage current does

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

not conform to 3.9.3, the insulation resistance does not conform to 3.4, the dielectric test produces a breakdown of the insulation or discloses inability to support the test voltage for the specified time, any circuit discontinuity is detected, or any instability in circuit or insulation resistance is observed.

4.5.12 Sustained operation test.

4.5.12.1 General. Three of the suppressor stacks (see table III) shall be given a sustained operation test to determine compliance with 3.10.

4.5.12.2 Description of the test. The sustained operation test shall consist of operation for 1025 ± 25 hours with a surge voltage pulse of 20 μs duration and a 60 Hz repetition rate which produces 0.5 ampere peak per square inch of effect cell area. The first 100 hours shall be continuous duty operation followed by 185 ± 5 cycles, 5 hours on and 1 hour off, of intermittent operation. The peak values of both surge current and suppressor voltage, simultaneous values of ambient temperature and suppressor temperature, and total operating time shall be measured and recorded initially, at 1, 2, and 4 hours, and after each subsequent 24-hour interval until 100 operating hours are accumulated; then hourly for the first, fifth, and thirteenth cycles of intermittent operation followed by a set of mid-cycle readings each seventh day, except full cycle hourly readings for one cycle each third week (providing 5 cycles with hourly readings). After the thirteenth cycle, the reading times may be adjusted to avoid having to take readings on Saturdays, Sundays, and holidays but the actual date, cycle number, and times from cycle initiation shall be recorded. No more than 72 hours shall elapse between completion of the continuous duty portion of the operation and commencement of the intermittent portion of the open-circuit peak surge voltage of 3000 ± 150 volts at the terminals to which the suppressor connects. The circuit impedance shall be set to give a peak surge current through the suppressor of 0.50 ± 0.05 amperes per square inch of effective cell area. A set of measurements to be identified as initial values as soon after the start of the test as feasible shall be made and recorded. The operation shall continue and another set of measurements shall be made and recorded at the appropriate times. If the peak surge current is outside the specified tolerance, the circuit impedance shall be readjusted to bring it back to the specified value, and another set of measurements shall be made and recorded at that measurement interval. This shall be continued for the duration of the test or until failure occurs. Upon completion of the test run, the test specimens shall be returned to room ambient temperature and shall be subjected, within 72 hours after completion of the performance run, to post-performance test measurements in accordance with 4.5.11.7, except that the requirement with regard to effective ambient temperature of initial measurements shall not apply.

4.5.13 Rate of voltage rise. A series of surge voltage pulses of 200 ± 100 μs duration which produce through the suppressor at least 5 amperes of current per square inch of effective cell area after the initial surge has subsided and with rates of voltage rise of 20 ± 2 volts per microsecond (V/μs), 200 ± 20 V/μs, and 2000 ± 200 V/μs, respectively, shall be impressed upon the suppressor under test. Performance shall be monitored and documented by recording the instantaneous voltage time and current time characteristics of the suppressor displayed on an oscilloscope. This series of surges shall be repeated two times for a total of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

three iterations. The individual surges shall be sufficiently spaced to avoid build-up of cell operating temperature in the suppressor. Both the suppressor and the recordings of instantaneous voltage and current characteristics shall be examined for any evidence of sparking or other malfunctioning, and for the magnitudes and relationship of any detectable let-through voltage to follow-on voltage. Any such evidence or failure to comply in any respect with 3.9.5 shall constitute failure of the specimen.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.7).

5.1 Preservation-packaging, packing and marking. Suppressors shall be individually preserved-packaged level A or C, packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555 and the following:

(a) For level A preservation-packaging, method III shall apply in lieu of the methods cited in MIL-E-17555.

5.2 Cushioning, dunnage, and wrapping materials.

5.2.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation/stowage.

5.2.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Selenium surge suppressor cells and stacks covered by this specification are intended for use in electric and electronic equipment for Naval use.

6.1.1 Nonpolarized suppressors. The nonpolarized suppressors are intended for use in a.c. circuits.

6.1.2 Polarized suppressors. The polarized suppressors are intended for use in d.c. circuits or in pulsed circuits where polarity is unidirectional.

6.1.3 Code C, commercial finish. The code C, commercial finish, is intended for use only on suppressors used in equipment which will be installed in air-conditioned spaces where there is no direct ingress of raw air from outside the ship.

6.1.4 Code H, military finish. The code H, military finish, is intended for use on suppressors used in equipment which may be installed in open deck areas, in the ship's superstructure, or in enclosed deck areas in or near the air stream from a ventilation system which discharges raw air brought in from outside the ship. The equipment enclosure and location in the ship should be such as will protect the suppressor from exposure to water, spray, or mist.

6.1.5 Code J, potted or embedded forms. The code J, potted or embedded forms, is intended for use only on certain small size (see 3.13.3) suppressors used in equipment under the same limitations as the code H finish (see 6.1.4).

6.1.6 Code M, marine finish. The code M, marine finish, is intended for use on suppressors used in equipment which will be installed in enclosed shipspaces in locations not in or near the discharge air stream of ventilating systems bringing in raw air from outside the ship. The code M finish requirements do not assure as high a degree of protection as those for codes H and J, but more than those for code C.

6.1.7 Code X, special finishes. The intended use for code X, special finish, will be as specified on the applicable specification control drawing. A code X finish is to be used only where one of the other types is not suitable.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of suppressor required (see 1.2 and 1.2.1).
(c) Type of finish coating required (see 3.13 and 4.5.11.5).
(d) Special requirements and tests necessary for the specific application (see table VI).
(e) Levels of preservation-packaging and packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisons of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
3.15 through 3.15.3	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - Contractor Drawing number - Contractor
4.4.4	Report, test/ inspection	UDI-T-23473	------

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data where submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24602 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

6.4 Definitions.

6.4.1 Selenium surge suppressor cell. A selenium surge suppressor cell is a polycrystalline semiconductor device having an asymmetrical voltage-current characteristic. The reverse portion of this characteristic exhibits its rapid change in slope at a higher voltage than does the same feature of the forward portion of the voltage-current characteristic. The cell functions as a surge suppressor when it is operated in the reverse portion of the voltage-current characteristic.

6.4.2 Selenium surge suppressor stack. A selenium surge suppressor stack is an integral assembly of one or more selenium surge suppressor cells with terminal connections and mounting provisions.

6.4.3 Polarized selenium surge suppressor stack. A polarized selenium surge suppressor stack is a stack in which each circuit element contains either a single cell or multiple cells in series-aiding so that the circuit element exhibits an asymmetrical voltage-current characteristic.

6.4.4 Nonpolarized selenium surge suppressor stack. A nonpolarized selenium surge suppressor stack is a stack in which each circuit element contains cell-pairs in series opposition so that the circuit element exhibits an approximately symmetrical voltage-current characteristic.

6.4.5 Selenium surge suppressor circuit assembly. A selenium surge suppressor circuit assembly is the combination (in one assembly) of one or more surge suppressor stacks consisting of all the suppressor circuit elements used in one complete suppressor circuit.

6.4.6 Electrode. An electrode is an electrical and mechanical contact to a region of a device.

6.4.7 Terminal. A terminal is the externally available point of connection to one or more electrodes.

6.4.8 Positive terminal. The positive terminal of a selenium surge suppressor stack or assembly is that terminal to which the surge current flows from the external circuit when the suppressor functions to suppress a voltage surge.

6.4.9 Negative terminal. The negative terminal of a selenium surge suppressor stack or assembly is that terminal from which the surge current flows to the external circuit when the suppressor functions to suppress a voltage surge.

6.4.10 A.c. terminal. The a.c. terminal of a selenium surge suppressor stack or assembly is one which receives bi-directional current flow in the quiescent state of the suppressor, and may receive current flow of either direction when the suppressor functions to suppress a voltage surge depending upon the polarity of the surge.

6.4.11 Form factor. The form factory of a wave form is the ratio of the rms value of the wave to the average value.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

6.4.12 Effective cell area. The effective cell area of a selenium surge suppressor cell is that nominal area of the cell which is available to carry the surge current.

6.4.13 Surge current. The surge current of a selenium surge suppressor (cell, stack, or assembly) is the current that flows through the suppressor when it is functioning to suppress a voltage surge.

6.4.14 Peak surge current rating. The peak surge current rating of a selenium surge suppressor (cell, stack, or assembly) is the maximum instantaneous current which the suppressor is rated to carry under specified conditions when it is functioning to suppress a voltage surge.

6.4.15 Steady-state leakage current. The steady-state leakage current is the current which flows through a surge suppressor (cell, stack, or assembly) when a voltage of value corresponding to the rated steady-state voltage is impressed upon the suppressor under rated conditions.

6.4.16 Steady-state voltage rating. The steady-state voltage rating of a selenium surge suppressor (cell, stack, or assembly) is the maximum voltage at which the suppressor is rated for continuous duty under specified conditions.

6.4.17 Clamping voltage. The clamping voltage of a selenium surge suppressor (cell, stack, or assembly) is the maximum instantaneous voltage drop across the suppressor resulting from the flow of rated peak surge current.

6.4.18 Clamping voltage rating. The clamping voltge rating of a selenium surge suppressor (cell, stack, or assembly) is that value of clamping voltage at which the manufacturer rates the suppressor such that at the required quality level, this rated value will not be exceeded by the product line.

6.4.19 Sparking voltage. The sparking voltage of a selenium surge suppressor (cell, stack, or assembly) is the instantaneous voltage impressed upon the suppressor at which dielectric failure occurs and a visible arc can be observed.

6.4.20 Sparking voltage rating. The sparking voltage rating of a selenium surge suppressor (cell, stack, or assembly) is that value assigned by the manufacturer which, at the required quality level, the sparking voltage of all specimens of the product line will lie.

6.4.21 Suppression voltage. The suppresion voltage of a selenium surge suppressor (cell, stack, or assembly) is the maximum instantaneous voltage that occurs across the suppressor during the suppression of a voltage surge (or passage of a current surge).

6.4.21.1 Let-through voltage. When the variation of instantaneous voltage over time tends to peak during the rise to maximum value of the applied surge voltage, this peak value is termed the let-through voltage.

6.4.21.2 Follow-on voltage. The more stable portion of a suppressed surge voltage which follows the let-through voltage is called the follow-on voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

6.4.22 Repetitive peak voltage. The repetitive peak voltage is the maximum instantaneous value of the suppressor voltage, including all repetitive transient voltages but excluding all nonrepetitive transient voltages, which occurs across a surge suppressor.

6.4.23 Nonrepetitive peak voltage. The nonrepetitive peak voltage is the maximum instantaneous value of the suppressor voltage including all nonrepetitive transient voltages but excluding all repetitive transient voltages, which occurs across a surge suppressor.

6.4.24 Forming. Forming is the electrical or thermal treatment, or both, of a selenium suppressor for the purpose of increasing the steady-state voltage capabilities and stabilizing its electrical characteristics.

6.4.25 Basic rating. The basic rating of a selenium surge suppressor (cell, stack, or assembly) is the maximum steady-state voltage (a.c. rms or d.c. average), the maximum nonsparking voltage, the maximum clamping voltage, and the maximum surge current at which the manufacturer rates the surge suppressor conforming to this specification for operation with convection cooling in the specified (see 3.6) ambient temperature. These rating factors are, respectively, the steady-state voltage rating, the sparking voltage rating, the clamping voltage rating, and the peak surge current rating. In this specification, reference to maximum nonrepetitive surge implies a time between such surges of at least 30 minutes.

6.5 Manufacturer's code symbols. The manufacturer's code symbols are contained in the Navy Code of Manufacturer's Names of the Master Identification Section of the Catalog of Navy Material and may be obtained from NAVSEA.

6.6 The following data concerning the suppressor stack or assembly and its application should be included on the equipment drawing in which it is installed:

(a) Manufacturer of suppressor.
(b) Manufacturer's type and catalog number of suppressor.
(c) Applicable Government specification and type designation.
(d) Applicable tabulated detail drawing (see 3.15.2) and piece numbers, or the specification control drawing number (see 3.15.3) if any classification category (see 1.2) is "special".
(e) Maximum and minimum operating temperature.
(f) Worst case electrical stresses as follows:
    (1) Nonrepetitive surge current.
    (2) Unsuppressed value of the potentially available nonrepetitive peak voltage.
    (3) Rate of voltage rise.
    (4) Power dissipation during surge and effective duration.
    (5) Sustained overvoltage (a.c. rms or d.c. average, as applicable - 30 seconds or longer duration).
    (6) Repetitive peak transient voltage and duration (less than 30 seconds).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

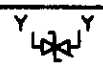
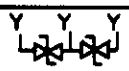
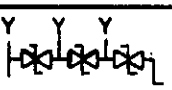
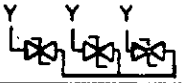
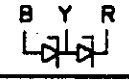
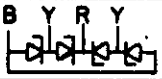
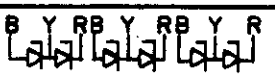
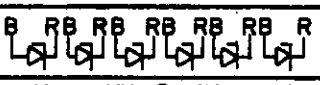
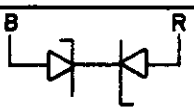
(7) Suppression voltage (suppressed value of voltage surge).

(g) Nominal steady-state voltage (a.c. rms or d.c. average; as applicable).

6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 6130-N193)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

Code	Circuit description	Schematic diagram
N1H	Non-polarized, single-element, half wave	
N2S	Non-polarized, two-element, center-tap	
N3R	Non-polarized, three-element, ring (delta)	
N3S	Non-polarized, three-element, star (wye)	
N4B	Non-polarized, four-element, bridge	
N6B	Non-polarized, six-element, bridge	
N6H	Non-polarized, six-element, all insulated	
P1H	Polarized, single-element, half-wave	
P2S	Polarized, two-element, center-tap, positive common	
P2D	Polarized, two-element, doubler	
P4B	Polarized, four-element, bridge	
P6B	Polarized, six-element, bridge	
P6D	Polarized, six-element, doubler	
P6H	Polarized, six-element, all insulated	
P6S	Polarized, six-element, star, positive common	
C	Combination polarized and regulatory, single-element, suppressor	 line surge discharge

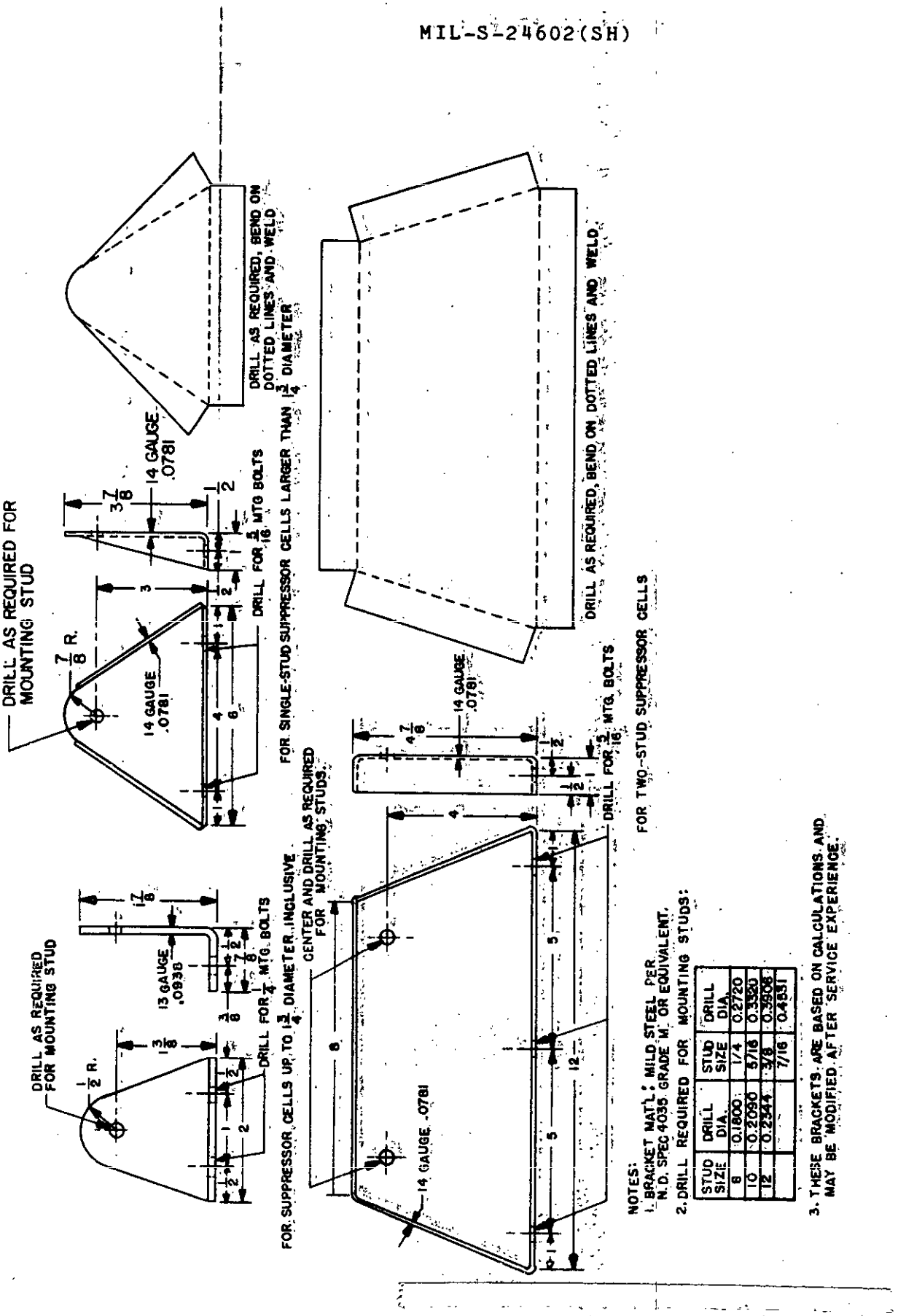
Color code and markings

Symbol	Color	Description	Marking
B	Black	Negative	—
R	Red	Positive	+
Y	Yellow	A.c.	A.c.

SH 11748

FIGURE 1. Circuits and connection arrangements for selenium surge suppressor stacks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.



STUD SIZE	DRILL DIA.	STUD SIZE	DRILL DIA.
8	0.1800	1/4	0.2720
10	0.2090	5/16	0.3320
12	0.2344	3/8	0.3908
		7/16	0.4531

FIGURE 2. Standard brackets for mounting suppressor stacks for shock test.

Source: https://assist.dla.mil - Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Sea System Command (SEA 3112)
Department of the Navy
Washington, DC 20362

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2025-10-19T23:14Z
Check the source to verify that this is the current version before use.